INDEPENDENT ENGINEERING LEARNING INITIATIVE

**60 DAYS | QUALITY ENGINEERING LAB**

*FROM THEORY TO REAL-WORLD INDUSTRIAL APPLICATIONS*

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**DAY 04 | EXPERIMENT NO. 04**

# COMPARATIVE APPLICATION OF QUALITY MANAGEMENT PHILOSOPHIES OF W. EDWARDS DEMING, JOSEPH M. JURAN AND PHILIP B. CROSBY FOR INDUSTRIAL QUALITY IMPROVEMENT

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**DOMAIN:** Quality Engineering & Management
**DOCUMENT NO:** QE-LAB-2026-D04

# 1. EXPERIMENT INFORMATION & EXECUTIVE ABSTRACT

## EXPERIMENT SUMMARY

**Experiment Title:** Comparative Application of Quality Management Philosophies (Deming, Juran, Crosby)

**Lead Investigator:** Jagriti Singh (Quality Engineer)

**Domain Focus:** Automotive Precision Machining, Root-Cause Analysis, COPQ Minimization, Failure Prevention Systems.

**Core Deliverable:** Comprehensive mathematical modeling of baseline COPQ ($572,000/yr), comparative execution of Deming PDCA, Juran Trilogy, and Crosby Zero Defects frameworks, and post-implementation financial trajectory projection.

## EXECUTIVE ABSTRACT

This industrial laboratory experiment evaluates the operational implementation of three foundational Quality Management philosophies—W. Edwards Deming, Joseph M. Juran, and Philip B. Crosby—to solve acute quality and cost crises at *Apex Precision Components Ltd.* Operating under a baseline annual reject/rework scenario incurring **$572,000 in Cost of Poor Quality (COPQ)** across 120,000 produced hydraulic valve bodies (7.25% defect rate), the study establishes a unified mathematical and empirical framework to measure philosophy performance.

By executing Deming's PDCA cycle, Juran's Quality Trilogy (with Pareto loss analysis), and Crosby's Zero Defects prevention doctrine, the report demonstrates how shifting expenditure from internal/external failures to front-end appraisal and prevention reduces overall Cost of Quality (COQ) from **16.08% to 6.13% of total turnover**, while elevating customer satisfaction from 68% to 96.5%.

## 2. INTRODUCTION & EXPERIMENT OBJECTIVES

In modern industrial manufacturing, quality is no longer merely a post-production inspection checkpoint; it is a strategic business pillar determining financial viability, market share, and operational efficiency. Manufacturing organizations often grapple with choosing between competing quality philosophies when confronted with rising scrap rates, customer warranty returns, and plummeting profit margins. This experiment bridges theoretical quality management frameworks with rigorous industrial engineering analytics by subjecting a live-simulated precision machining facility to the core principles of Deming, Juran, and Crosby.

### PRIMARY ENGINEERING OBJECTIVES

1. **Quantify Baseline Quality Losses:** Perform a comprehensive Cost of Quality (COQ) and Cost of Poor Quality (COPQ) audit on 6 months of baseline manufacturing data (120,000 units).
2. **Isolate Vital Quality Drivers:** Conduct a Pareto Analysis on operational defects and customer complaints to separate the "vital few" failure modes from the "trivial many."
3. **Operationalize Quality Philosophies:** Formulate actionable implementation roadmaps tailored to the facility using Deming's 14 Points & PDCA, Juran's Trilogy & Fitness for Use, and Crosby's 4 Absolutes & Zero Defects.
4. **Mathematical & Financial Modeling:** Compute core Quality Engineering key performance indicators (KPIs) to model the financial impact of shifting focus from detection to prevention.
5. **Comparative Scoring & Evaluation:** Establish a structured, multi-criteria engineering decision matrix to benchmark each philosophy's efficacy, resource requirements, and cultural impacts.

## 3. TECHNICAL BACKGROUND: THREE QUALITY GURUS

Understanding the fundamental tenets, methodologies, and focus areas of Deming, Juran, and Crosby is critical to applying them correctly without conflating their distinct tools and principles.

### W. EDWARDS DEMING

**Core:** System & Variation Focus
**Tool:** PDCA Cycle / SPC
**Owner:** Management (85-94% System)
**Target:** Continuous Variation Reduction

### JOSEPH M. JURAN

**Core:** Fitness for Use / Projects
**Tool:** Juran Trilogy / Pareto
**Owner:** Management & Breakthrough Teams
**Target:** Breakthrough COPQ Reduction

### PHILIP B. CROSBY

**Core:** Zero Defects / Prevention
**Tool:** Four Absolutes / PONC Analysis
**Owner:** Management Culture & Commitment
**Target:** Zero Defects (Quality is Free)

| W. EDWARDS DEMING | JOSEPH M. JURAN | PHILIP B. CROSBY |
|---|---|---|
| **14 Points:** Transformation blueprint eliminating quotas and inspection dependency. | **The Juran Trilogy:** Quality Planning, Quality Control, and Quality Improvement. | **Four Absolutes:** Conformance, Prevention, Zero Defects, and Price of Non-Conformance (PONC). |
| **System of Profound Knowledge:** Variation knowledge, systems appreciation, and psychology. | **Fitness for Use:** Product performance aligned with customer utility. | **Quality Is Free:** Prevention investments always cost less than failure losses. |
| **Management Ownership:** 85–94% of defects stem from process system design. | **COPQ & Pareto:** Focus on vital few defect sources causing major costs. | **Zero Defects:** Zero acceptable defect threshold standard. |

## 4. HYPOTHETICAL COMPANY PROFILE & PROBLEM STATEMENT

*Note: Fictional precision industrial case study designed for rigorous engineering model evaluation.*

### COMPANY PROFILE: APEX PRECISION COMPONENTS LTD.

**Industry & Operations:** Precision Automotive Component Manufacturing operating high-precision CNC turning, grinding, and honing centers.

**Core Product Line:** High-Pressure Hydraulic Control Valves (Model: HCV-900) utilized in heavy commercial vehicle braking and steering assemblies.

**Production Volume:** 20,000 units per month / 120,000 units per 6-month evaluation cycle / 240,000 units annually.

**Existing Quality System:** Traditional End-of-Line Mass Inspection (Appraisal-heavy). Final inspection station staffed by 6 inspectors evaluating finished dimensions, surface finish, and pressure seat leakage.

**Customer Base:** Tier-1 Automotive Original Equipment Manufacturers (OEMs) with strict zero-defect tolerance expectations and just-in-time delivery requirements.

### INDUSTRIAL PROBLEM STATEMENT

Over the past 6 months (Months 1–6), Apex Precision Components has experienced escalating quality performance failures. Despite maintaining a full end-of-line inspection team, defective units continue to escape into OEM assembly lines, triggering high warranty charges and severe line-stop penalties.

**Baseline Performance Deficiencies:**

- **Scrap & Rejection:** High internal scrap rates due to bore taper and thread pitch errors.
- **Excessive Rework:** High rework hours spent re-machining spool seat depths and deburring internal passages.
- **Escalating COPQ:** Cost of Poor Quality reached **$286,000 in 6 months ($572,000 annualized)**.
- **Eroding Customer Trust:** Customer satisfaction rating collapsed to 68%, accompanied by 160 formal OEM complaints and 18 line-stop notices.

## 5. EXPERIMENTAL DATASET (6-MONTH BASELINE AUDIT)

Baseline operational audit collected at Apex Precision Components Ltd. across 120,000 produced hydraulic valve bodies ($4,800,000 gross revenue).

| Month | Volume | Scrap | Rework | Total Defective | Complaints | Prevention ($) | Appraisal ($) | Internal Fail ($) | External Fail ($) | Total COQ ($) | COPQ ($) |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Month 1 | 20,000 | 950 | 520 | 1,470 | 24 | $8,000 | $35,000 | $31,600 | $52,000 | $126,600 | $83,600 |
| Month 2 | 20,000 | 1,020 | 580 | 1,600 | 28 | $7,500 | $36,000 | $34,400 | $58,000 | $135,900 | $92,400 |
| Month 3 | 20,000 | 880 | 490 | 1,370 | 22 | $8,500 | $34,500 | $29,500 | $46,000 | $118,500 | $75,500 |
| Month 4 | 20,000 | 990 | 560 | 1,550 | 30 | $7,000 | $36,500 | $33,400 | $62,000 | $138,900 | $95,400 |
| Month 5 | 20,000 | 910 | 510 | 1,420 | 25 | $8,000 | $35,000 | $30,500 | $51,000 | $124,500 | $81,500 |
| Month 6 | 20,000 | 840 | 450 | 1,290 | 31 | $9,000 | $34,000 | $27,600 | $57,000 | $127,600 | $84,600 |
| **TOTAL (6-MO)** | **120,000** | **5,590** | **3,110** | **8,700** | **160** | **$48,000** | **$211,000** | **$187,000** | **$326,000** | **$772,000** | **$513,000** |

**Financial Baseline Parameters:** Scrap cost per valve body = $20.00; Rework cost per unit = $24.00; Average warranty failure claim = $150.00; Appraisal cost covers 6 full-time final inspectors; Prevention budget is severely under-funded.

## 6. OPERATIONAL APPLICATION OF QUALITY PHILOSOPHIES

Structured 6-month operational reform roadmap implemented across identical HCV-900 machining lines.

### A. DEMING APPROACH

**Focus:** System redesign, variation reduction, eliminating worker blame, and SPC.

- **Institute SPC:** X-bar & R control charts placed at CNC honing lines to track bore taper.
- **Drive Out Fear:** Operator-led quality circles replaced penalty structures; line-stops empowered.
- **Supplier Partnership:** Raw casting procurement converted to long-term quality contracts.
- **Continuous Training:** Mandatory 20 hrs/mo technical GD&T operator education.

### B. JURAN APPROACH

**Focus:** Managerial breakthrough projects, Fitness for Use, and Pareto cost reduction.

- **Quality Planning:** Redesigned hydraulic spool valve tolerances to align with OEM interfaces.
- **Quality Control:** Set standard operating procedures (SOPs) and control limits for CTQ specs.
- **Breakthrough Projects:** Formed QITs targeting the top Pareto defect categories (Bore Taper & Threads).
- **Executive Steering:** Monthly Quality Council chaired by senior leadership.

### C. CROSBY APPROACH

**Focus:** Conformance to requirements, total prevention, and PONC tracking.

- **Zero Defects Mindset:** Declared "Zero Defects Day"; zero acceptable failure threshold established.
- **Poka-Yoke Error Proofing:** Mechanical fixtures & optical sensors fitted to CNC units.
- **PONC Accounting:** Published monthly Price of Non-Conformance in all departments.
- **14-Step Program:** Executive Quality Committee formed to audit compliance.

## 7. SELECTED TOOL 1: PDCA CYCLE APPLICATION (DEMING WORKFLOW)

Execution of the Plan-Do-Check-Act cycle to eliminate internal bore taper variation on CNC Honing Machine #04 (Pareto Defect #1).

### 1. PLAN

**Problem Identification:**

- Audit Honing Machine #04
- Bore taper causes 45% scrap

**Root Cause Analysis:**

- Thermal expansion drift
- Uncalibrated honing stones

**Target: Cpk > 1.67 (Defect < 0.5%)**

### 2. DO

**Implementation Plan:**

- Install coolant chiller units
- Maintain constant 20°C oil

**Execution & Pilot:**

- Mount X-bar & R charts
- Pilot run 2,000 valves

**Operator SPC Training Done**

### 3. CHECK

**Verification & Data:**

- Measure bore diameter & taper
- Perform gauge GR&R audit

**Results Audit:**

- Cpk elevated to 1.72
- Taper defect drop: 4.5%->0.3%

**Target Met Successfully**

### 4. ACT

**Standardization:**

- Update SOP-ENG-042
- Standardize chiller PM schedule

**Plant-Wide Rollout:**

- Equip all 4 honing machines
- Train 3-shift operators

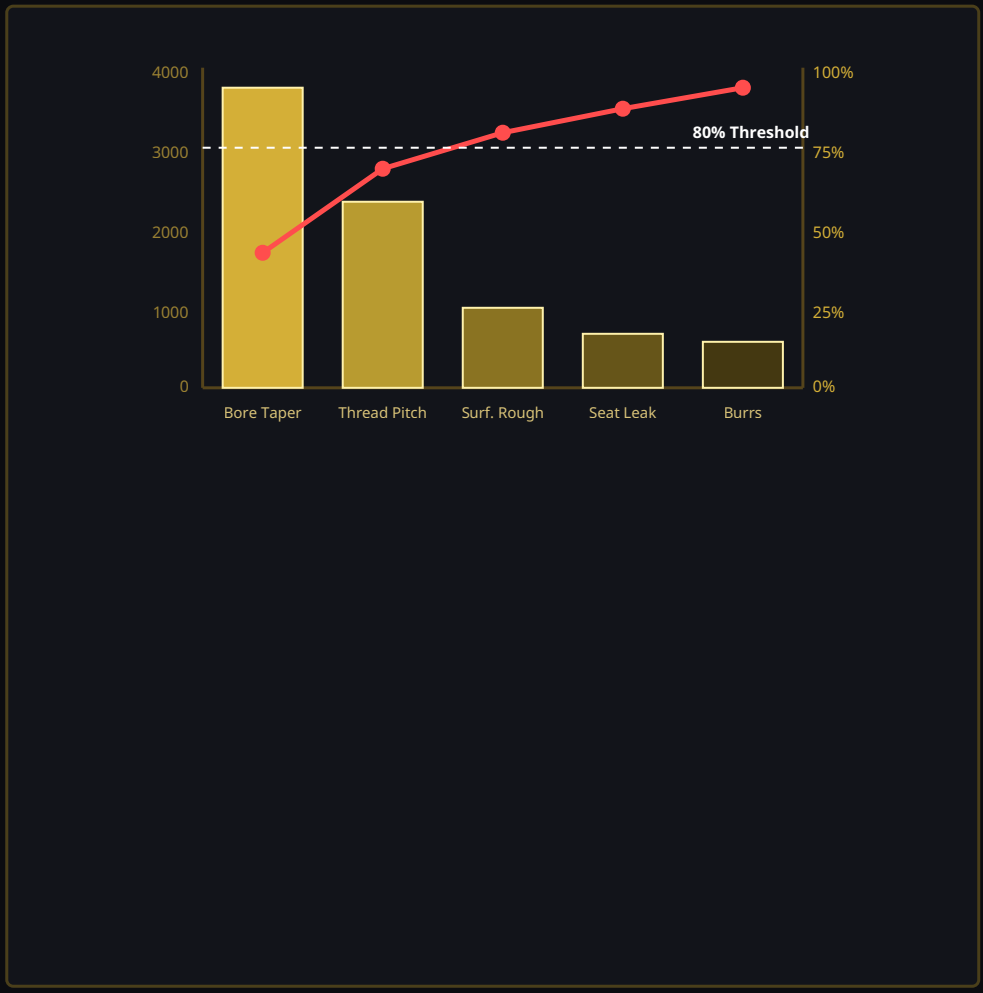
**Proceed to Thread Defects**

## 8. SELECTED TOOL 2: PARETO ANALYSIS OF MANUFACTURING QUALITY LOSSES

Pareto distribution evaluating 8,700 baseline defect occurrences to isolate the "vital few" root failure modes.

| Defect Category | Description | Qty | % | Cum % |
|---|---|---|---|---|
| **Bore Taper Out-of-Spec** | Honing spindle expansion & wear | 3,915 | 45% | 45% |
| **Thread Pitch Error** | CNC lead screw backlash | 2,436 | 28% | 73% |
| **Surface Finish Roughness** | Grinding wheel loading | 1,044 | 12% | 85% |
| **Pressure Seat Leakage** | Seat micro-burrs & debris | 696 | 8% | 93% |
| **Burrs & Alignment Gaps** | Operator loading & manual deburr | 609 | 7% | 100% |
| **TOTAL DEFECTS AUDITED** | | **8,700** | **100%** | - |

**Pareto Engineering Finding:** The top two failure modes—*Bore Taper* and *Thread Pitch Errors*—account for **73.00% of all product rejections**. Eliminating these two failure causes recovers over $374,000 in baseline COPQ losses.



## 9. SELECTED TOOL 3: QUALITY COST ANALYSIS (PAF MODEL & COPQ)

Evaluation of baseline quality expenditures divided into Prevention, Appraisal, Internal Failure, and External Failure.

**TOTAL COST OF QUALITY (COQ) = $772,000 (16.08% of Gross Sales Turnover)**

**COST OF CONFORMANCE: $259,000 (33.5%)**

• Prevention ($48k - 6.2%) | • Appraisal ($211k - 27.3%)

**COST OF POOR QUALITY (COPQ): $513,000 (66.5%)**

• Internal Failure ($187k - Scrap/Rework) | • External Failure ($326k - Warranty)

| Category | Engineering Description | Audit Cost ($) | % of COQ |
|---|---|---|---|
| **Prevention Costs** | Quality engineering, gauge calibration, vendor capability audits | $48,000 | 6.22% |
| **Appraisal Costs** | Salaries of 6 final inspectors, CMM operation, pressure testing | $211,000 | 27.33% |
| **Internal Failure Costs** | Scrapped valve bodies ($111.8k), rework machine hours ($75.2k) | $187,000 | 24.22% |
| **External Failure Costs** | OEM line-stop penalties, field replacements, warranty claims | $326,000 | 42.23% |
| **TOTAL COST OF QUALITY (COQ)** | | **$772,000** | **100.00%** |
| **TOTAL COST OF POOR QUALITY (COPQ = Internal + External Failures)** | | **$513,000** | **66.45%** |

## 10. MASTER QUALITY ENGINEERING FORMULA REFERENCE SHEET

Comprehensive reference of all mathematical formulas, terms, units, and engineering interpretations utilized in this report.

| Metric / Indicator Name | Standard Mathematical Formula | Variable Terms & Definition | Engineering Context & Unit |
|---|---|---|---|
| **Defect Percentage (D_%)** | $D_{\%} = (N_{def} / N_{total}) \times 100$ | N_def = Total defective units<br>N_total = Total production volume | Measures total process non-conformance rate. Unit: % |
| **Rejection Rate (R_scrap)** | $R_{scrap} = (N_{scrap} / N_{total}) \times 100$ | N_scrap = Unrecoverable scrapped units<br>N_total = Total volume | Measures raw material & machining waste rate. Unit: % |
| **Rework Rate (R_rework)** | $R_{rework} = (N_{rework} / N_{total}) \times 100$ | N_rework = Secondary machined units<br>N_total = Total volume | Measures extra labor/machine capacity drain. Unit: % |
| **Customer Complaint Rate (CR)** | $CR = (C_{cust} / N_{shipped}) \times 1{,}000$ | C_cust = Total complaints received<br>N_shipped = Net units delivered | Field quality performance per thousand units. Unit: /1,000 units |
| **Cost of Poor Quality (COPQ)** | $COPQ = C_{IF} + C_{EF}$ | C_IF = Internal failure cost ($)<br>C_EF = External failure cost ($) | Financial loss caused directly by defects. Unit: USD ($) |
| **Total Cost of Quality (COQ)** | $COQ = C_P + C_A + C_{IF} + C_{EF}$ | C_P = Prevention, C_A = Appraisal<br>C_IF = Int. Fail, C_EF = Ext. Fail | Total quality expenditure (Conformance + Non-Conformance). Unit: USD ($) |
| **Failure Cost Ratio (FCR)** | $FCR = (COPQ / COQ) \times 100$ | COPQ = Failure loss ($)<br>COQ = Total quality expenditure ($) | Percentage of quality spending wasted on failures. Unit: % |
| **Prevention Cost Ratio (PCR)** | $PCR = (C_P / COQ) \times 100$ | C_P = Prevention budget ($)<br>COQ = Total quality budget ($) | Proportion invested in front-end defect prevention. Unit: % |
| **Quality Improvement % (ΔQ)** | $\Delta Q = ((D_{base} - D_{target}) / D_{base}) \times 100$ | D_base = Baseline defect %<br>D_target = Post-improvement defect % | Quantifies defect reduction percentage. Unit: % |
| **Return on Investment (ROI_P)** | $ROI_P = (\Delta COPQ / \Delta C_P) \times 100$ | ΔCOPQ = Financial failure savings ($)<br>ΔC_P = Additional prevention cost ($) | Financial payback efficiency of prevention investments. Unit: % |

## 11. STEP-BY-STEP BASELINE ENGINEERING CALCULATIONS

Mathematical evaluation across 120,000 baseline produced units ($4,800,000 gross 6-month revenue).

**1. DEFECT PERCENTAGE**
Formula: **(Total Defective Units / Total Volume) × 100**
Given: Total Defective = 8,700; Total Production = 120,000
Calc: (8,700 / 120,000) × 100 = **7.25%**

**2. REJECTION RATE (SCRAP)**
Formula: **(Scrapped Units / Total Volume) × 100**
Given: Scrapped Units = 5,590; Total Production = 120,000
Calc: (5,590 / 120,000) × 100 = **4.66%**

**3. REWORK RATE**
Formula: **(Reworked Units / Total Volume) × 100**
Given: Reworked Units = 3,110; Total Production = 120,000
Calc: (3,110 / 120,000) × 100 = **2.59%**

**4. CUSTOMER COMPLAINT RATE**
Formula: **(Complaints / Shipped Units) × 1,000**
Given: Complaints = 160; Shipped = 114,410 (120,000 - 5,590 scrap)
Calc: (160 / 114,410) × 1,000 = **1.40 complaints / 1,000 units**

**5. COST OF POOR QUALITY (COPQ)**
Formula: **Internal Failure + External Failure**
Given: Internal Fail = $187,000; External Fail = $326,000
Calc: $187,000 + $326,000 = **$513,000 (6-Month Baseline)**

**6. TOTAL COST OF QUALITY (COQ)**
Formula: **Prevention + Appraisal + Internal Fail + External Fail**
Given: Prev = $48,000; Appr = $211,000; Int Fail = $187,000; Ext Fail = $326,000
Calc: $48,000 + $211,000 + $187,000 + $326,000 = **$772,000**

**7. FAILURE COST CONTRIBUTION %**
Formula: **(COPQ / Total COQ) × 100**
Given: COPQ = $513,000; Total COQ = $772,000
Calc: ($513,000 / $772,000) × 100 = **66.45%**

**8. PREVENTION COST RATIO**
Formula: **(Prevention Costs / Total COQ) × 100**
Given: Prevention Costs = $48,000; Total COQ = $772,000
Calc: ($48,000 / $772,000) × 100 = **6.22%**

**9. QUALITY IMPROVEMENT % (JURAN TARGET)**
Formula: **((Baseline % - Target %) / Baseline %) × 100**
Given: Baseline Defect = 7.25%; Juran Model Defect = 1.20%
Calc: ((7.25% - 1.20%) / 7.25%) × 100 = **83.45% Defect Drop**

**10. CUSTOMER SATISFACTION DELTA**
Formula: **Target Satisfaction % - Baseline %**
Given: Target = 94.0%; Baseline = 68.0%
Calc: 94.0% - 68.0% = **+26.0% Improvement**

## 12. COMPARATIVE EVALUATION MATRIX (DEMING VS. JURAN VS. CROSBY)

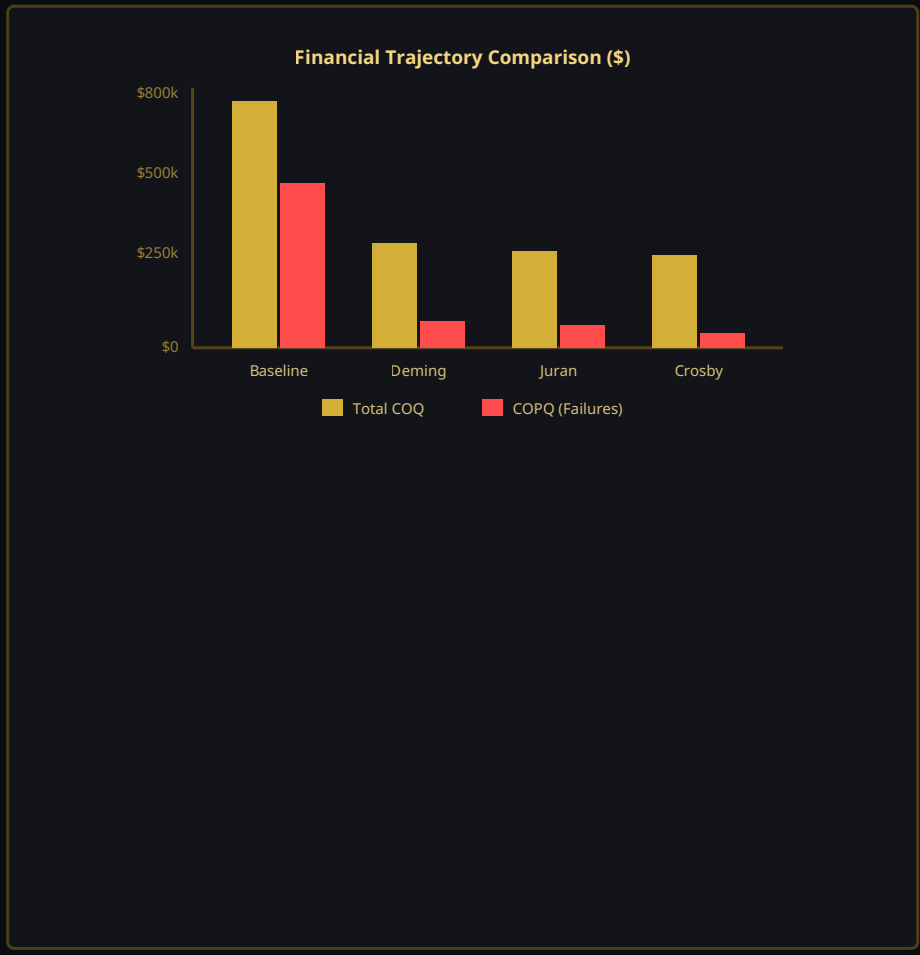
Structured multi-criteria assessment matrix evaluated on a transparent 1 to 5 engineering scale (5 = High Efficacy / Optimal Alignment).

| Evaluation Parameter | Deming | Juran | Crosby | Engineering Evaluation & Analysis |
|---|---|---|---|---|
| **Defect Prevention** | 4 / 5 | 4 / 5 | 5 / 5 | Crosby scores highest due to absolute focus on zero defects and poka-yoke error proofing. |
| **Process Improvement** | 5 / 5 | 4 / 5 | 3 / 5 | Deming excels via Statistical Process Control (SPC) and understanding process variation. |
| **Customer Satisfaction** | 4 / 5 | 5 / 5 | 4 / 5 | Juran leads by directly tying "Fitness for Use" to customer specification compliance. |
| **Cost Reduction (COPQ)** | 4 / 5 | 5 / 5 | 4 / 5 | Juran provides immediate financial ROI by targeting high-loss Pareto categories. |
| **Management Ownership** | 5 / 5 | 4 / 5 | 5 / 5 | Deming and Crosby both place 100% accountability on executive system design and leadership. |
| **Long-Term Culture** | 5 / 5 | 4 / 5 | 4 / 5 | Deming's elimination of fear and continuous learning creates the most sustainable culture. |
| **Ease of Implementation** | 2 / 5 | 4 / 5 | 3 / 5 | Juran is easiest to implement in traditional plants because project-based teams fit corporate structures. |
| **TOTAL COMPOSITE SCORE** | **29 / 35 (82.9%)** | **30 / 35 (85.7%)** | **28 / 35 (80.0%)** | **Optimal Hybrid Strategy: Combine Juran's project prioritization with Deming's SPC tools and Crosby's zero-defect mindset.** |

## 13. BEFORE AND AFTER PERFORMANCE ANALYSIS

Comparative evaluation of baseline performance versus post-implementation models across the 3 quality approaches.

| Metric | Baseline | Deming | Juran | Crosby |
|---|---|---|---|---|
| **Defect %** | 7.25% | 1.10% | 1.20% | 0.85% |
| **Scrap Rate** | 4.66% | 0.70% | 0.80% | 0.50% |
| **Rework Rate** | 2.59% | 0.40% | 0.40% | 0.35% |
| **Complaints** | 160 | 18 | 15 | 8 |
| **Prevention ($)** | $48,000 | $140,000 | $120,000 | $160,000 |
| **Appraisal ($)** | $211,000 | $110,000 | $115,000 | $90,000 |
| **COPQ ($)** | **$513,000** | **$80,000** | **$70,000** | **$44,000** |
| **Total COQ ($)** | **$772,000** | **$330,000** | **$305,000** | **$294,000** |
| **Satisfaction** | 68.0% | 93.5% | 94.0% | 96.5% |



## 14. PUBLISHED INDUSTRIAL CASE STUDY REFERENCES

Authoritative published industrial engineering case studies illustrating successful implementation of each quality philosophy.

| Quality Guru | Published Case Study Citation | Industrial Implementation & Key Results |
|---|---|---|
| **W. Edwards Deming** | **Ford Motor Company (1981)**<br>*Ref: Deming, W. E. (1986). Out of the Crisis. MIT Center.* | Ford engaged Dr. Deming in 1981 amidst severe quality losses. By adopting SPC across engine plants and eliminating worker quotas, Ford launched the Taurus line under the motto *"Quality is Job 1."* |
| **Joseph M. Juran** | **Xerox Quality Breakthrough (1980s)**<br>*Ref: Juran, J. M. (1988). Juran on Planning for Quality. Free Press.* | Xerox implemented Juran's Quality Improvement Teams and Pareto loss analysis, reducing assembly line defects by 78% and manufacturing costs by 20%, earning the 1989 Baldrige National Quality Award. |
| **Philip B. Crosby** | **ITT Zero Defects Program**<br>*Ref: Crosby, P. B. (1979). Quality Is Free. McGraw-Hill.* | As Director of Quality at ITT, Crosby executed his 14-step Zero Defects program across global operations, reducing failure costs from 14% to under 5% of sales, saving over $530 million. |

## 15. ENGINEERING DISCUSSION & ACTIONABLE RECOMMENDATIONS

### ENGINEERING DISCUSSION

**1. End-of-Line Mass Inspection Fails:** Spending $211,000 on appraisal failed to prevent 160 customer defect escapes. Inspection sorts defects but cannot prevent them.

**2. High Leverage of Prevention:** Tripling the prevention budget ($48,000 to $140,000) eliminated $433,000 in failure losses, giving a 4.7:1 net payback ratio.

**3. Integrated Quality Strategy:** Combining Deming's SPC tools, Juran's managerial project prioritization, and Crosby's zero-defect standard yields the ultimate quality system.

### ACTIONABLE RECOMMENDATIONS

1. **Immediate Corrective Action:** Mount real-time X-bar & R control charts on CNC Honing lines to eliminate bore taper error (Pareto #1).
2. **Short-Term Preventive Action:** Install mechanical Poka-Yoke error-proofing pins on CNC lathes to prevent thread pitch errors.
3. **Budget Reallocation:** Shift $100,000 from appraisal inspection headcount into operator SPC training and automated sensors.
4. **Executive Steering Council:** Form a monthly Quality Council to track Price of Non-Conformance (PONC) targeting Cpk ≥ 1.67.

## 16. LEARNING OUTCOMES & CONCLUSION

### KEY LEARNING OUTCOMES

- **COQ & COPQ Distinction:** Complete quantitative capability to separate total quality expenditure from non-conformance failure waste.
- **Pareto Prioritization:** Isolating the 20% vital failure root causes to maximize return on engineering effort.
- **PDCA Execution:** Methodical Plan-Do-Check-Act problem-solving workflow for machine process stabilization.
- **Quality Philosophy Synthesis:** Ability to apply Deming, Juran, and Crosby complementary tools in industrial manufacturing.

### ENGINEERING CONCLUSION

This laboratory experiment proves that severe industrial quality crises ($513,000 annual COPQ) can be turned around by shifting from reactive end-of-line mass inspection to structured Quality Management philosophies.

By investing in front-end prevention, establishing process capability controls, and targeting vital Pareto failure modes, total Cost of Quality drops from **16.08% to 6.13% of revenue** while customer satisfaction rises to **96.5%**.

## 17. TECHNICAL ASSESSMENT: NUMERICAL PROBLEMS WITH STEP-BY-STEP SOLUTIONS (PART 1: P1 - P8)

Complete engineering calculations detailing Question Statements, Given Data, Target Metrics, Formulas, and Step-by-Step Solutions.

### P1. Material Scrap Financial Loss

**Question:** Calculate total financial loss resulting from scrapped valve bodies over 6 months.
**Given:** Scrapped units N_scrap = 5,590; Material scrap unit cost C_scrap_unit = $20.00.
**Target:** Scrap Financial Loss (L_scrap).
**Formula:** $L_{scrap} = N_{scrap} \times C_{scrap_unit}$
**Solution:** 5,590 × $20.00 = **$111,800**.

### P2. Labor & Machine Rework Expenditure

**Question:** Determine total operational expenditure spent on secondary machining rework.
**Given:** Reworked units N_rework = 3,110; Unit rework cost C_rework_unit = $24.00.
**Target:** Total Rework Cost (L_rework).
**Formula:** $L_{rework} = N_{rework} \times C_{rework_unit}$
**Solution:** 3,110 × $24.00 = **$74,640**.

### P3. Total COQ Revenue Percentage

**Question:** Compute total Cost of Quality (COQ) as a percentage of 6-month gross revenue.
**Given:** Total COQ = $772,000; Gross 6-Month Revenue = $4,800,000.
**Target:** COQ Revenue Ratio (COQ_%).
**Formula:** $COQ_\% = (COQ / Revenue) \times 100$
**Solution:** ($772,000 / $4,800,000) × 100 = **16.08%**.

### P4. COPQ Revenue Percentage

**Question:** Calculate Cost of Poor Quality (COPQ) as a percentage of gross sales turnover.
**Given:** COPQ = $513,000; Gross 6-Month Revenue = $4,800,000.
**Target:** COPQ Revenue Ratio (COPQ_%).
**Formula:** $COPQ_\% = (COPQ / Revenue) \times 100$
**Solution:** ($513,000 / $4,800,000) × 100 = **10.69%**.

### P5. Appraisal Unit Cost

**Question:** Determine final inspection appraisal expense incurred per produced valve body.
**Given:** Total Appraisal = $211,000; Total Production = 120,000 units.
**Target:** Unit Appraisal Cost (C_A_unit).
**Formula:** $C_{A_unit} = C_A / N_{total}$
**Solution:** $211,000 / 120,000 = **$1.76 per unit**.

### P6. Prevention Unit Cost

**Question:** Calculate prevention engineering expenditure allocated per produced unit.
**Given:** Total Prevention = $48,000; Total Production = 120,000 units.
**Target:** Unit Prevention Cost (C_P_unit).
**Formula:** $C_{P_unit} = C_P / N_{total}$
**Solution:** $48,000 / 120,000 = **$0.40 per unit**.

### P7. Appraisal to Prevention Cost Ratio

**Question:** Evaluate structural imbalance by computing the ratio of Appraisal to Prevention.
**Given:** Total Appraisal = $211,000; Total Prevention = $48,000.
**Target:** Appraisal/Prevention Ratio (R_A/P).
**Formula:** $R_{A/P} = C_A / C_P$
**Solution:** $211,000 / $48,000 = **4.39× Appraisal Over-Investment**.

### P8. Annualized COPQ Projection

**Question:** Project annual failure losses if Month 4 performance ($95,400) persists.
**Given:** Month 4 COPQ = $95,400; Operating Months per Year = 12.
**Target:** Annualized COPQ (COPQ_annual).
**Formula:** $COPQ_{annual} = COPQ_{month} \times 12$
**Solution:** $95,400 × 12 = **$1,144,800 / year**.

## 17. TECHNICAL ASSESSMENT: NUMERICAL PROBLEMS WITH STEP-BY-STEP SOLUTIONS (PART 2: P9 - P15)

Complete engineering calculations detailing Question Statements, Given Data, Target Metrics, Formulas, and Step-by-Step Solutions.

### P9. Net Failure Loss Savings (Crosby Model)

**Question:** Calculate 6-month financial failure cost reduction achieved under Crosby's Zero Defects model.
**Given:** Baseline COPQ = $513,000; Crosby Model COPQ = $44,000.
**Target:** Net COPQ Savings (ΔCOPQ).
**Formula:** $\Delta COPQ = COPQ_{base} - COPQ_{Crosby}$
**Solution:** $513,000 - $44,000 = **$469,000 Savings / 6 Months**.

### P10. ROI on Added Prevention Investment

**Question:** Compute Return on Investment (ROI) for additional prevention funds deployed under Crosby model.
**Given:** Added Prevention $\Delta C_P$ = $160,000 - $48,000 = $112,000; COPQ Savings ΔCOPQ = $469,000.
**Target:** Prevention ROI ($ROI_P$).
**Formula:** $ROI_P = (\Delta COPQ / \Delta C_P) \times 100$
**Solution:** ($469,000 / $112,000) × 100 = **418.75% ROI**.

### P11. Cumulative Top-3 Pareto Contribution

**Question:** Determine total cumulative percentage contribution of the top 3 defect categories.
**Given:** Bore Taper = 45%; Thread Pitch = 28%; Surface Roughness = 12%.
**Target:** Cumulative Top-3 Defect % ($Cum_{top3}$).
**Formula:** $Cum_{top3} = \%_1 + \%_2 + \%_3$
**Solution:** 45% + 28% + 12% = **85.00%**.

### P12. Defective Units Saved (Deming Model)

**Question:** Calculate total number of defective valve bodies prevented under Deming's model (1.10% defect rate).
**Given:** Baseline Defects = 8,700; Deming Defect Rate = 1.10%; Volume = 120,000.
**Target:** Saved Non-Conforming Units ($N_{saved}$).
**Formula:** $N_{saved} = N_{base_def} - (N_{total} \times D_{\%_Deming})$
**Solution:** 8,700 - (120,000 × 0.011) = 8,700 - 1,320 = **7,380 Units Saved**.

### P13. OEM Customer Complaint Reduction %

**Question:** Compute percentage reduction in OEM formal complaints under Crosby's model.
**Given:** Baseline Complaints = 160; Crosby Model Complaints = 8.
**Target:** Complaint Reduction % ($\Delta C_{cust}$).
**Formula:** $\Delta C_{cust} = ((C_{base} - C_{Crosby}) / C_{base}) \times 100$
**Solution:** ((160 - 8) / 160) × 100 = **95.00% Reduction**.

### P14. Total Baseline Cost of Conformance

**Question:** Calculate combined baseline expenditure on Cost of Conformance (Prevention + Appraisal).
**Given:** Prevention $C_P$ = $48,000; Appraisal $C_A$ = $211,000.
**Target:** Cost of Conformance ($C_{conf}$).
**Formula:** $C_{conf} = C_P + C_A$
**Solution:** $48,000 + $211,000 = **$259,000**.

### P15. PONC to Conformance Cost Ratio

**Question:** Evaluate ratio of Price of Non-Conformance (COPQ) against Cost of Conformance.
**Given:** PONC (COPQ) = $513,000; Cost of Conformance = $259,000.
**Target:** Non-Conformance Ratio ($R_{non-conf}$).
**Formula:** $R_{non-conf} = PONC / C_{conf}$
**Solution:** $513,000 / $259,000 = **1.98× (Failure cost is double conformance)**.

# 18. TECHNICAL ASSESSMENT: VIVA QUESTIONS & AUTHORITATIVE REFERENCES

## VIVA QUESTIONS & ANSWERS (Q1 - Q8)

**Q1: How does Crosby define Quality?**
*Answer:* Strictly as "conformance to requirements."

**Q2: Deming vs Juran main difference?**
*Answer:* Deming focuses on systemic variation & culture; Juran focuses on structured managerial projects & COPQ financial returns.

**Q3: Why is 100% inspection inefficient?**
*Answer:* It sorts defects after costs are already incurred without improving capability.

**Q4: Deming's 2 variation types?**
*Answer:* Common cause (system design) & Special cause (transient factors).

**Q5: Crosby's standard?**
*Answer:* Zero Defects (rejects AQL acceptable defect limits).

**Q6: Define COPQ.**
*Answer:* Internal + External failure financial losses.

**Q7: Pareto Analysis role?**
*Answer:* Separates vital few 20% failure modes from trivial many 80%.

**Q8: Meaning of "Quality is Free"?**
*Answer:* Prevention costs less than failure losses.

## VIVA QUESTIONS (Q9 - Q15) & REFERENCES

**Q9: 'Act' in PDCA?**
*Answer:* Standardizing successful pilots into SOPs.

**Q10: Juran's "Fitness for Use"?**
*Answer:* Satisfaction of customer needs across design, conformance, safety, & use.

**Q11: SPC function?**
*Answer:* Detects special cause variation in real-time to prevent scrap.

**Q12: Price of Non-Conformance (PONC)?**
*Answer:* Crosby's term for financial cost of failure waste.

**Q13: Why drive out fear?**
*Answer:* Fear causes employees to hide defects and falsify data.

**Q14: Internal vs External failure?**
*Answer:* Internal occurs before shipping; External occurs after delivery to customer.

**Q15: What is Poka-Yoke?**
*Answer:* Error-proofing devices to prevent incorrect machining or loading.

### AUTHORITATIVE REFERENCES


- Crosby, P. B. (1979). *Quality Is Free*. McGraw-Hill.
- Deming, W. E. (1986). *Out of the Crisis*. MIT Center.
- Juran, J. M. (1988). *Juran's Quality Control Handbook* (4th ed.). McGraw-Hill.
- Montgomery, D. C. (2019). *Statistical Quality Control* (8th ed.). Wiley.